DSSAT

Download → Installation →Workflow

DSSAT: Decision Support System for Agrotechnology Transfer

Prepared by/ Marwa Samy

Download

- https://dssat.net/main-download/

Main Download

Search...

Official DSSAT Version

DSSAT can be downloaded from this site. We gather some information from you and assign a unique serial number to your copy.

Download DSSAT v4.8.5

* DSSAT Version 4.8.5 released in December 1, 2024.

CRAFT

The **CGIAR Research Program on Climate Change, Agriculture and Food Security** (CCAFS) has initiated the development of a new personal computer-based decision support system for short-term and long-term yield forecasting, agricultural risk analysis associated with increasing extreme events and climate change impact studies. The CCAFS Regional Agricultural Forecasting Toolbox (CRAFT) includes the client application with a user-friendly interface and the database implementation. It is designed to use gridded data

In This Page

- Official DSSAT Version
- CRAFT

Request DSSAT

DSSAT is *Free of charge*!
Request to download your own copy today!

- DSSAT Version 4.8.5 **(New)**
  - Released on December 1, 2024

Download DSSAT v4.8.5

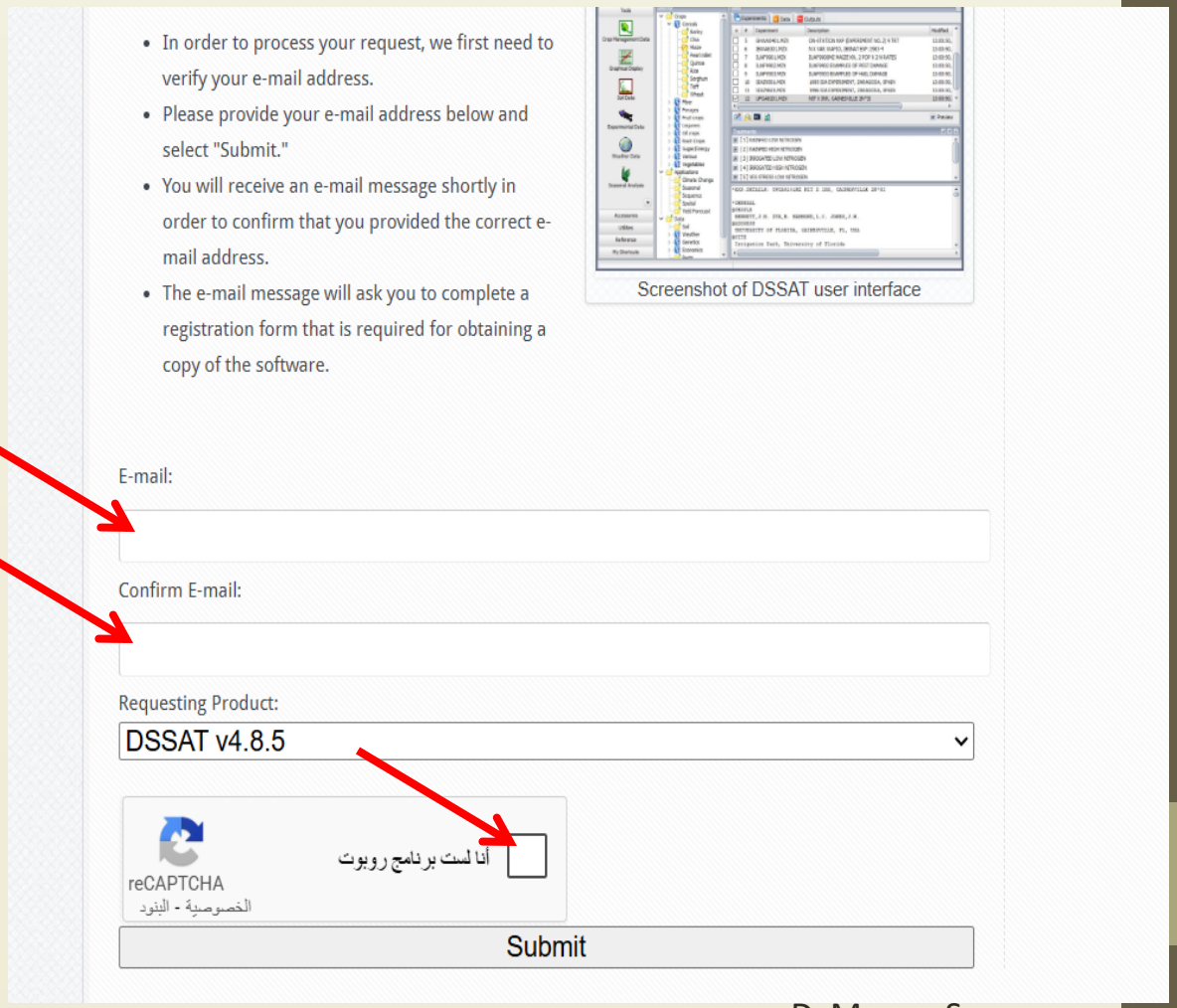
- Click here to see release notes
- Click here to download older versions of DSSAT

DSSAT

D. Marwa Samy

Get Serial Number

- To get your Serial Number to Optimization DSSAT, you have to write your e-mail and confirm you are not robot.
- Your Serial Number will be send via e-mail.



Get Serial Number

DF

DSSATFoundation<request@dssat.net>

To: You

Dear

To download the DSSAT installation files please use this link:
https://get.dssat.net/dssat-download-v4-8-5//?sk=48032501498

This is your DSSAT v4.8.5 Serial Number:

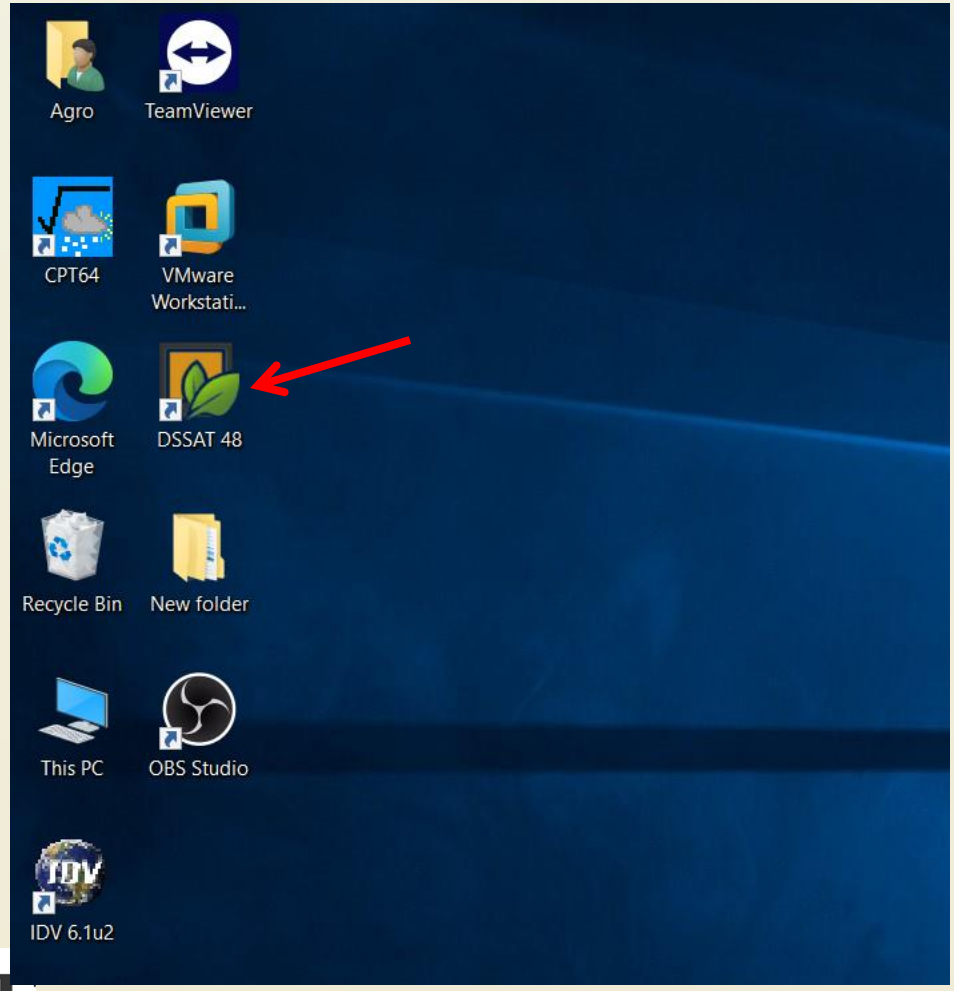
Please note that the size of download is large and that you will need a fast internet connection.

Regards,

The DSSAT Foundation

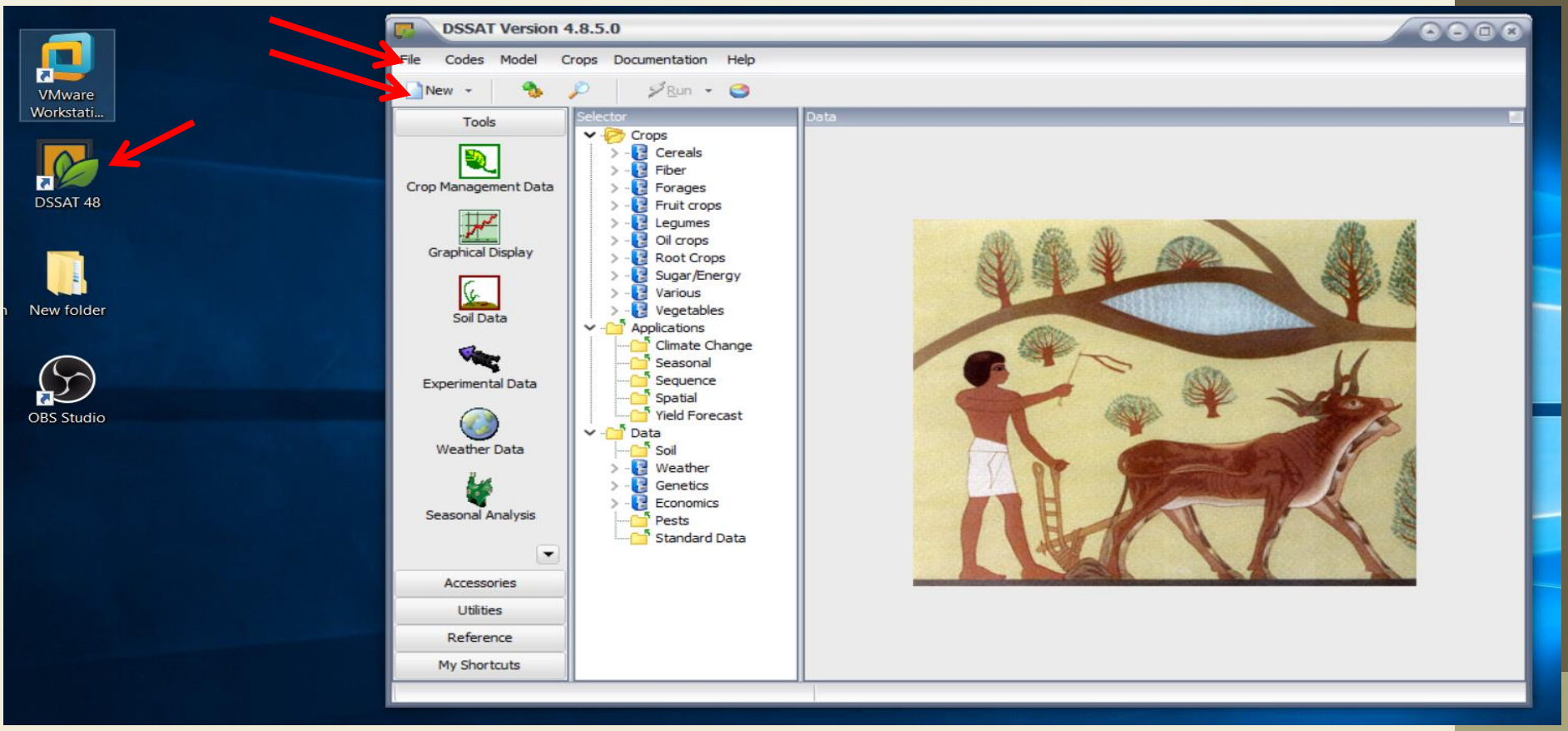
Reply Forward

Begin your journey



D. Marwa Samy

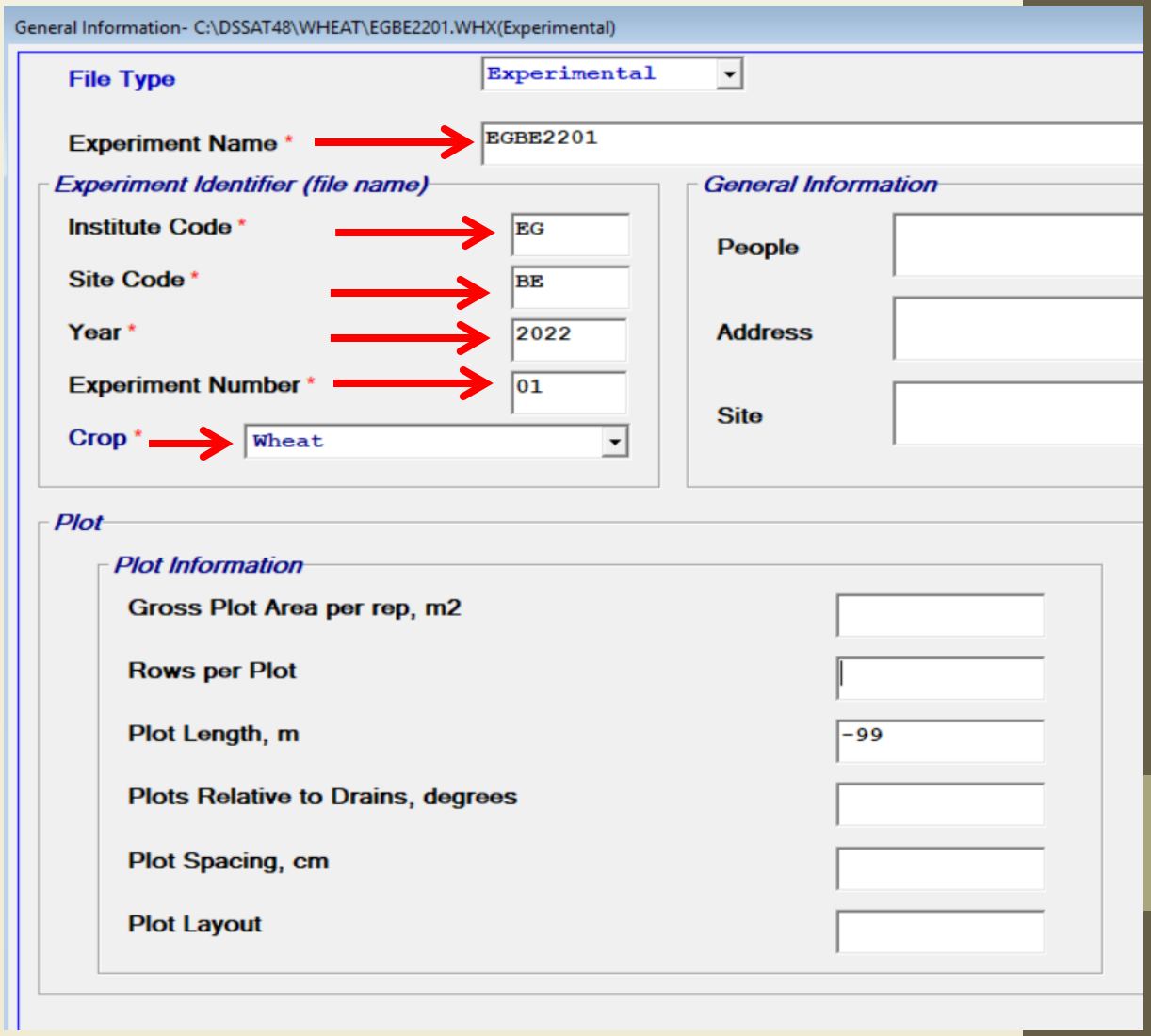
Optimization
1) Open Experiment data file



- **New**
- **Experiment**

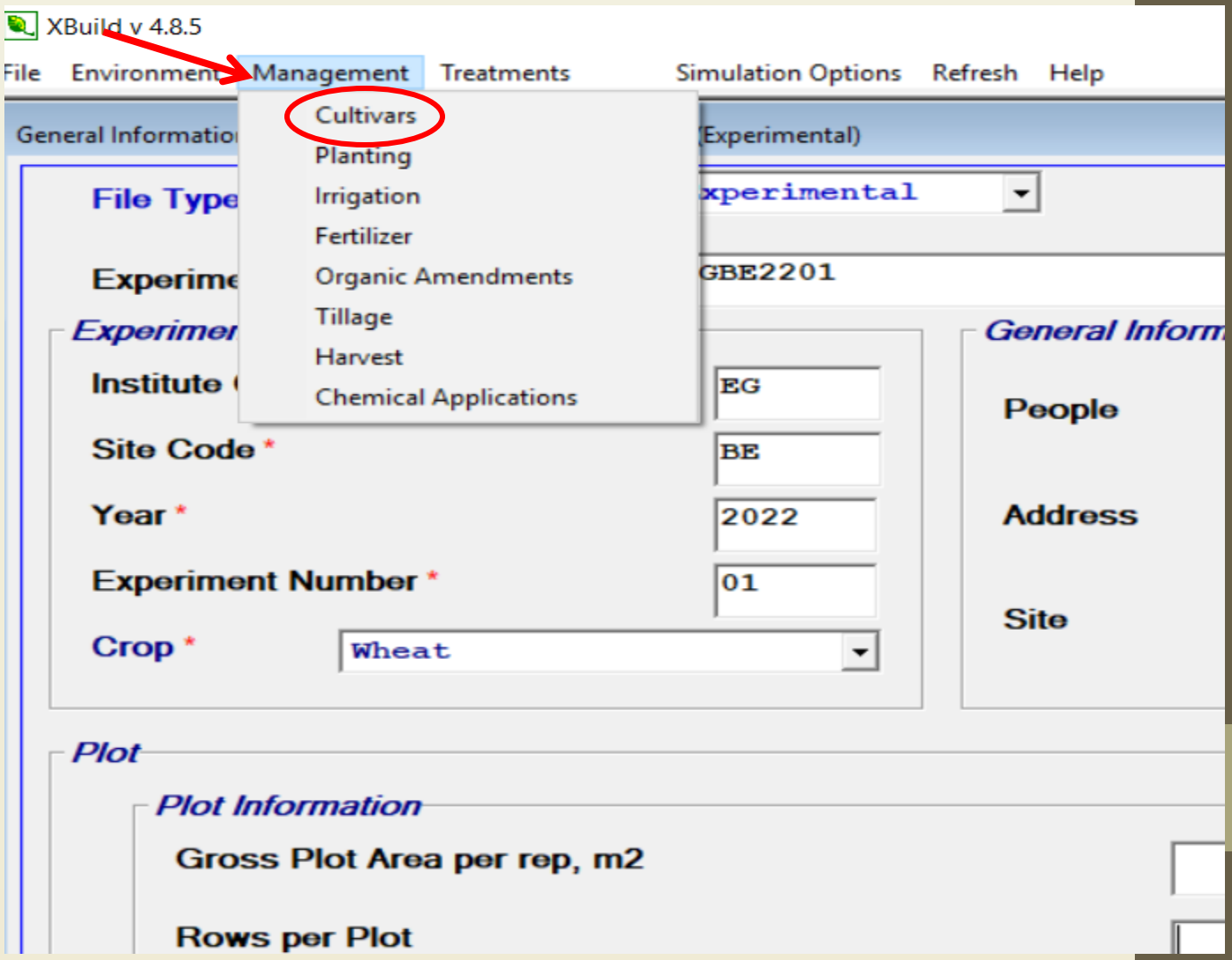
Optimization
2) General information

- **Institute code (2 digits)**
- **Site code (2 digits)**
- **Year (4 digits)**
- **Experiment No. (2 digits)**
- **Experiment name (Institute code + Site code + Year[3:4] + Experiment No.)**
- **Crop**

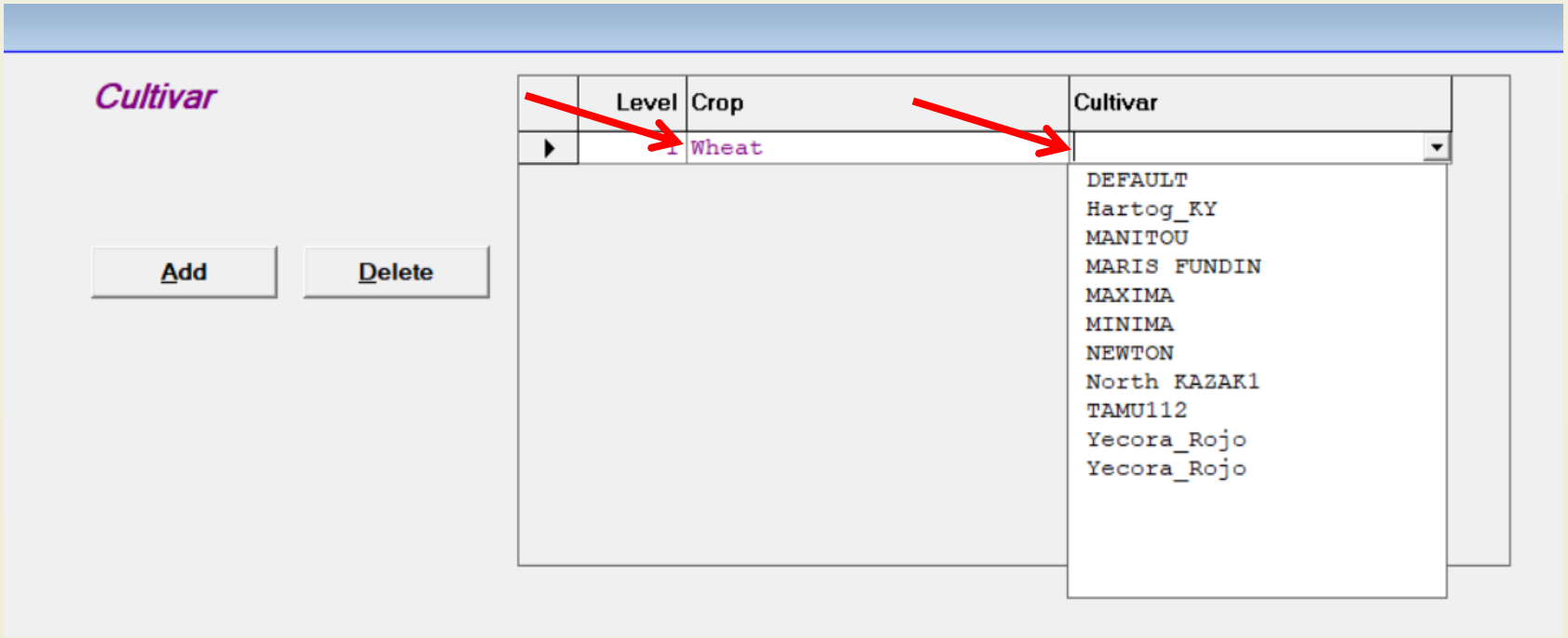


Optimization
3) Management → Cultivars

- تحديد سلالة النبات
- **Plant cultivar identification**



Optimization
3) Management → Cultivars



- **Determine crop**
- **Determine cultivar**
- **For information about cultivars → C:\DSSAT48\Genotype**

1) .CUL — Cultivar file

Phenology coefficients (e.g., thermal time to flowering/maturity)

Yield formation parameters

Grain filling duration

Sensitivity to photoperiod (often cultivar-level)

Genetic Difference

2) .ECO — Ecotype file

Environmental Adaptation

Response to photoperiod

Temperature adaptation ranges

Growth habit differences (spring vs winter wheat)

3) .SPE – Species file

Base, optimum, maximum temperatures

Radiation use efficiency

Carbon partitioning rules

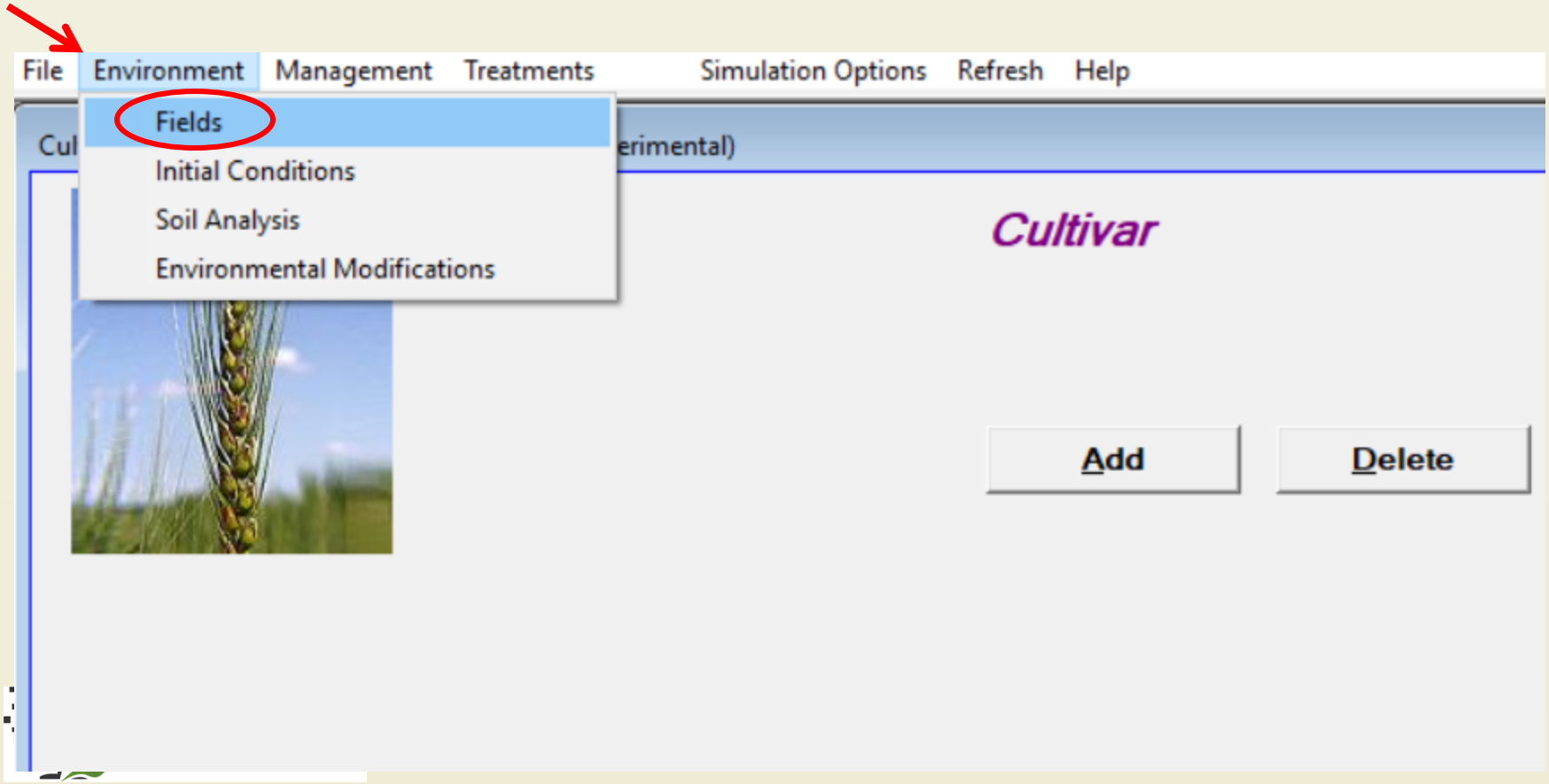
Root growth parameters

Stress response functions

Species level – Physiological constants

Optimization
4) Environment → Field

- **Weather and Soil files**

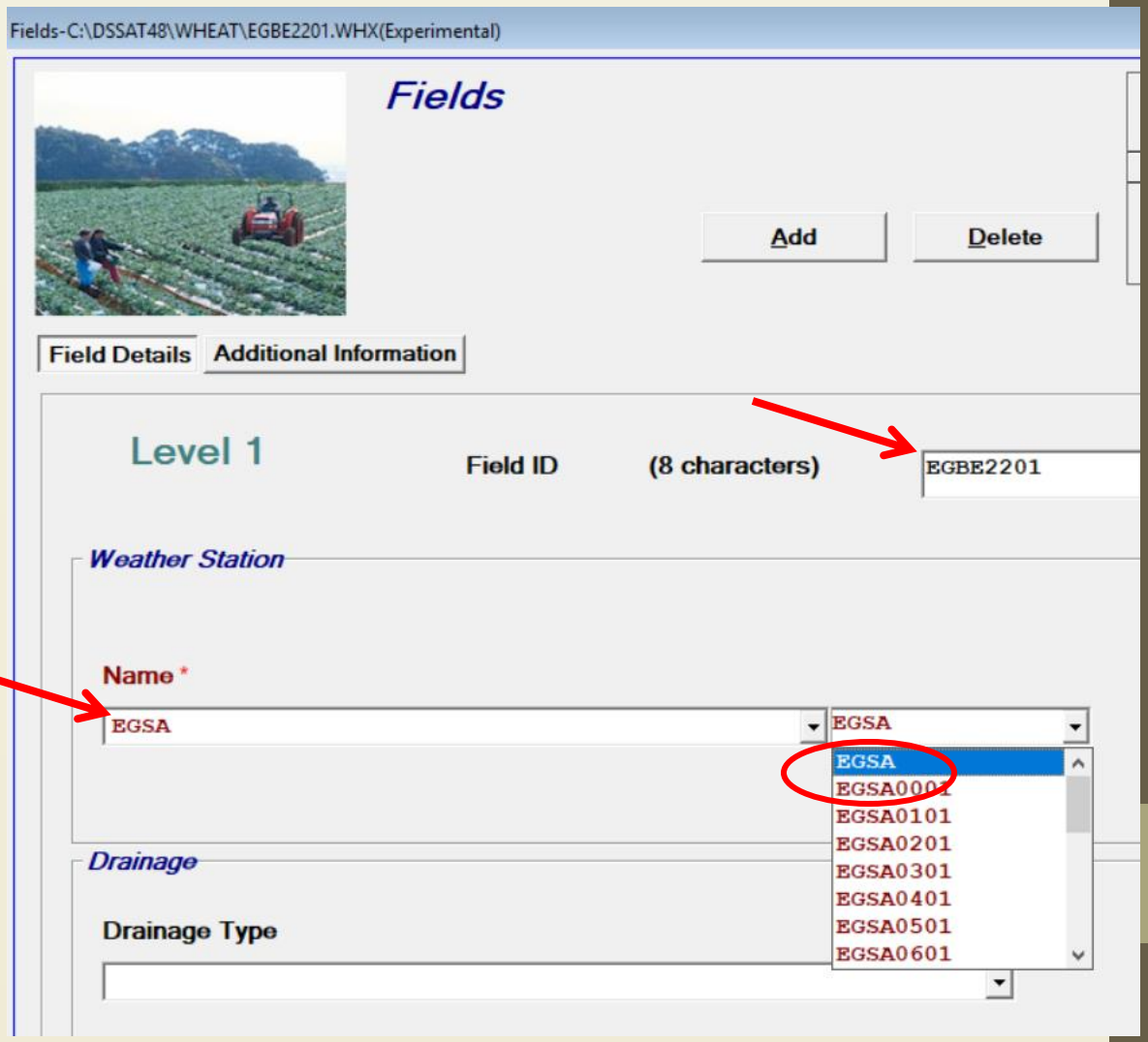


Optimization
4) Environment → Field

(a) **Enter Field ID**

(b) **Choose your weather file**

C:\DSSAT48\Weather



DSSAT

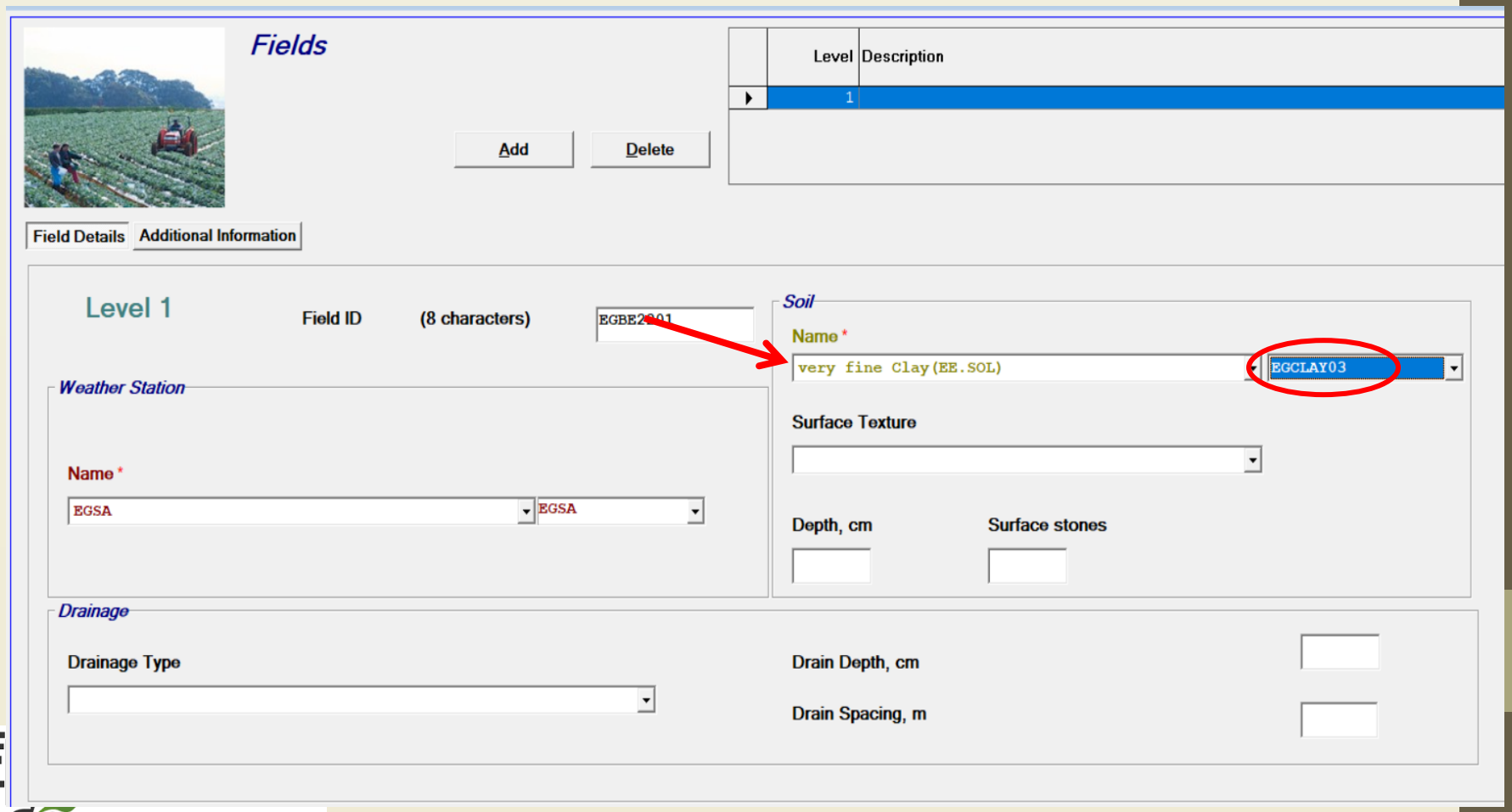
** Weather file **

```
$WEATHER DATA : EGSA

@ INSI      LAT     LONG  ELEV   TAV   AMP REFHT WNDHT
  EGSA   30.000   30.000   150  23.1  17.1 -99.0 -99.0
@  DATE  SRAD  TMAX  TMIN  RAIN  WIND
2013001  12.2  20.5   7.1   0.0   1.6
2013002  13.2  21.4   6.2   0.0   1.6
2013003  13.2  22.3   9.1   0.0   1.1
2013004  12.1  20.8   8.1   0.0   1.9
2013005  11.8  18.6   8.1   0.1   3.0
2013006  11.9  18.7   7.3   0.0   5.8
2013007   9.9  17.0   9.8   0.1   5.8
2013008   7.8  18.0   7.3   0.5   5.1
2013009   9.4  13.1   4.5   0.0   3.9
2013010   9.6  12.8   3.6   0.0   1.9
2013011  12.7  16.4   3.7   0.0   1.9
2013012  12.2  18.7   7.9   0.0   1.8
2013013  13.6  20.5   4.8   0.0   1.8
2013014  14.4  20.8   7.4   0.0   2.1
2013015  14.4  21.7   7.2   0.0   2.6
2013016   9.8  21.4   6.5   0.0   1.3
2013017  13.9  23.0   7.0   0.0   1.3
```

Optimization
4) Environment → Field

(C) Choose the soil file C:\DSSAT48\Soil



** Soil File **

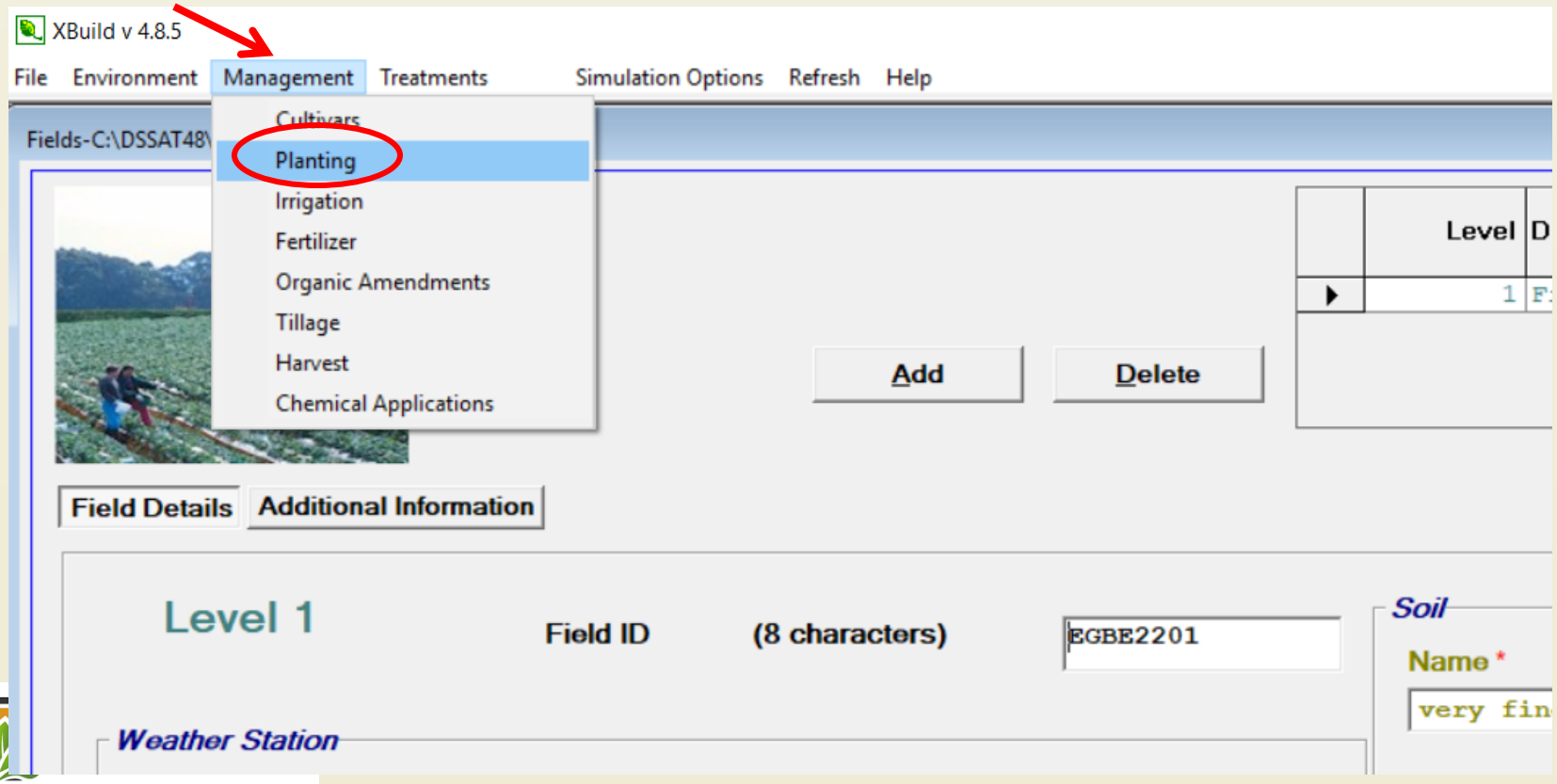
```
*SOILS: General DSSAT Soil Input File
*
*EGCLAY03     EC5            c     -99 very fine Clay
@SITE         COUNTRY          LAT     LONG SCS FAMILY
 Egypt        Sharqia        30.0   30.0 -99
@ SCOM  SALB  SLU1  SLDR  SLRO  SLNF  SLPF  SMHB  SMPX  SMKE
   -99  0.13   5.6   0.6    64   1     1      -99   -99   -99
@  SLB  SLMH  SLLL  SDUL  SSAT  SRGF  SSKS  SBDM  SLOC  SLCL  SLSI  SLCF  SLNI  SLHW  SLHB  SCEC  SADC
    15   -99  0.44  0.53  0.63  1     1.58  0.98  1.2    40    30   -99   -99   -99   -99   20   -99
    30   -99  0.44  0.53  0.63  0.7   1.58  0.98  1      40    30   -99   -99   -99   -99   18   -99
    50   -99  0.44  0.53  0.63  0.4   1.58  0.98  0.8    40    30   -99   -99   -99   -99   15   -99
! Derived from WOFOST EC5-very fine
```

For variable description **C:\DSSAT48\SOIL.CDE**

```
! Codes currently used for both detailed profile analysis and occasional
! analysis of the surface layers are listed. The soil analysis codes are
! also listed in the DATA.CDE file.
!
! The fields in the file are as follows:
!  CDE    The 'universal' code used to facilitate data interchange.
!  DESCRIPTION  A description of the code, with units.
!  SO     The source of the codes (IB=IBSNAT). Codes added by a user should be
!         referenced in this field and the name and address of the person adding
!         the code should be entered as a comment (ie.with a '!' in column 1)
!         below this note. This is important to ensure that information from
!         different workers can be easily integrated.

@CDE      DESCRIPTION
C         Clay
CACO3     CaCO3 content, g kg-1
CL        Clay loam
L         Loam
LAT       Latitude, degrees (decimals;+ve north)
LONG      Longitude, degrees (decimals;+ve east)
LS        Loamy sand
S         Sand
SABD      Bulk density, moist, g cm-3
SABL      Depth, base of layer, cm
SADAT     Analysis date, year + days from Jan. 1
SADC      Anion adsorption coefficient (reduced nitrate flow), cm3 (H2O) g [soil]-1
SAHB      pH in buffer
```

Optimization
5) Management → Planting



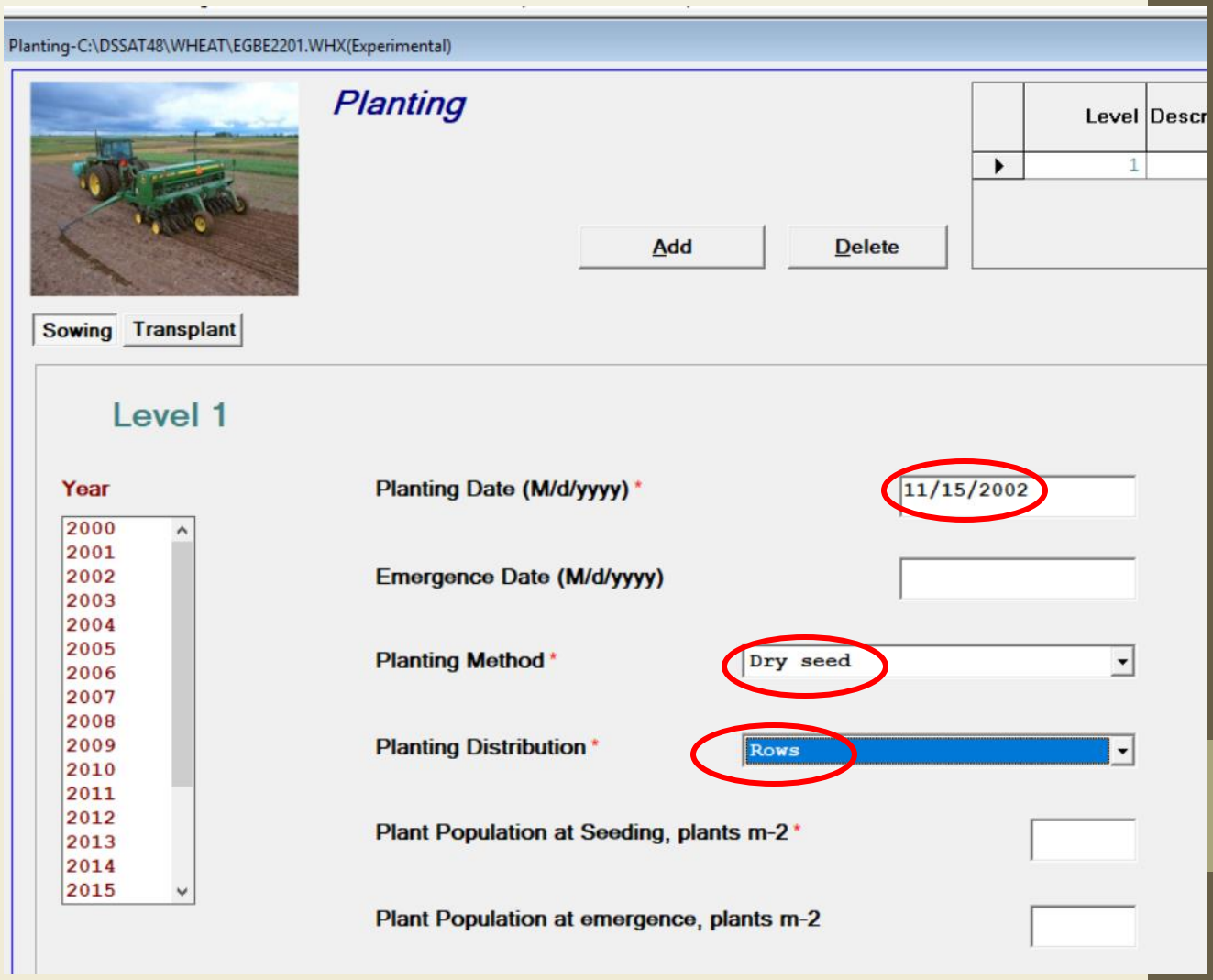
Optimization
5) Management → Planting

- **Define**

(a) **Planting date**

(b) **Planting method**

(c) **planting distribution**



Optimization
5) Management → Planting

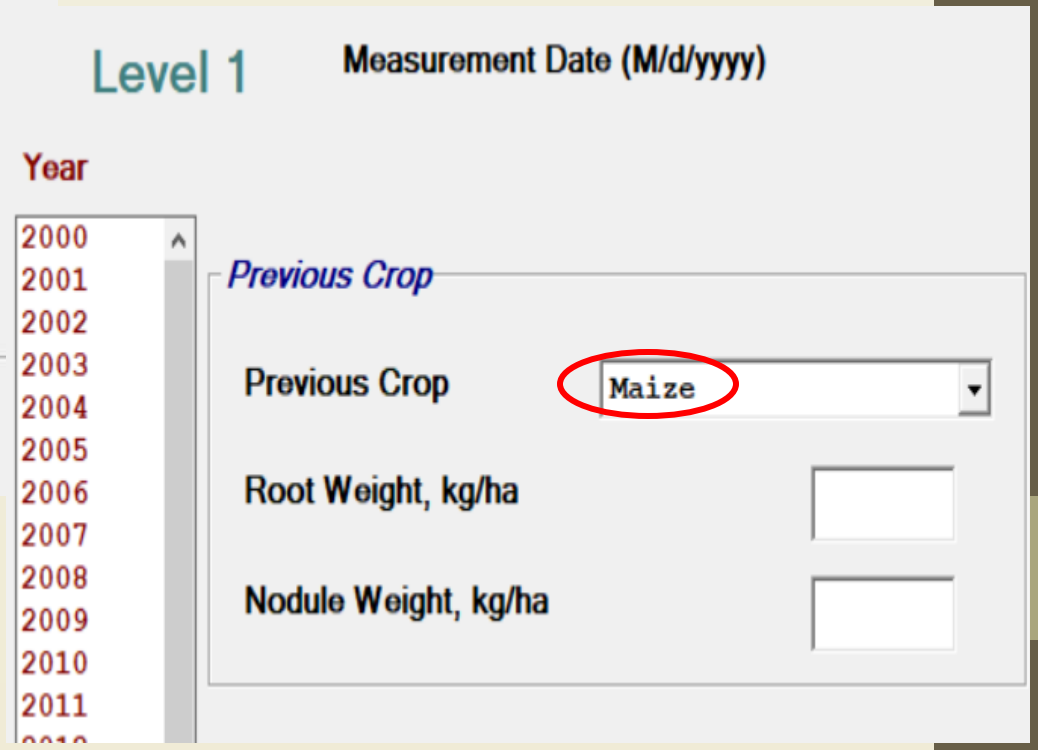
d) The spacing between rows (cm)
e) how many plants are present per square meter (Plant population of seeding)
f) Planting depth (cm)

Planting Distribution *	Rows	Row Spacing, cm *	15
Plant Population at Seeding, plants m-2 *	200	Row Direction, degrees from North *	
Plant Population at emergence, plants m-2		Planting Depth, cm *	4

DSSAT

D. Marwa Samy

Optimization
6) Environment → Initial conditions



Optimization
6) Environment → Initial conditions

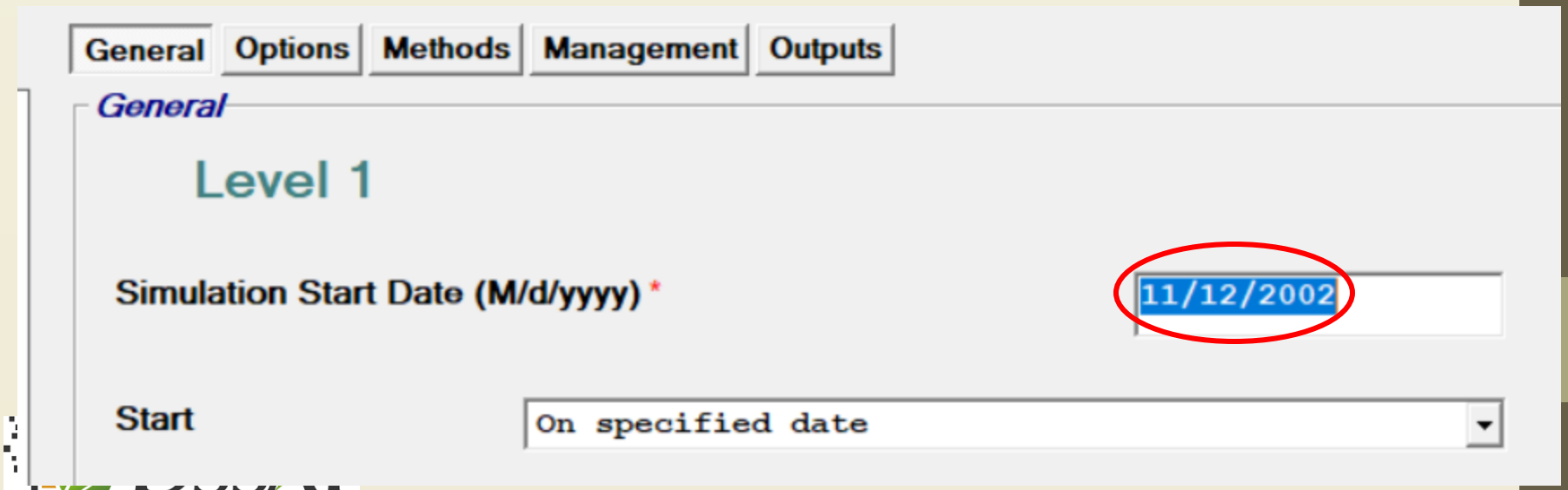
- **Measurement date is the same as the planting date**

DSSAT

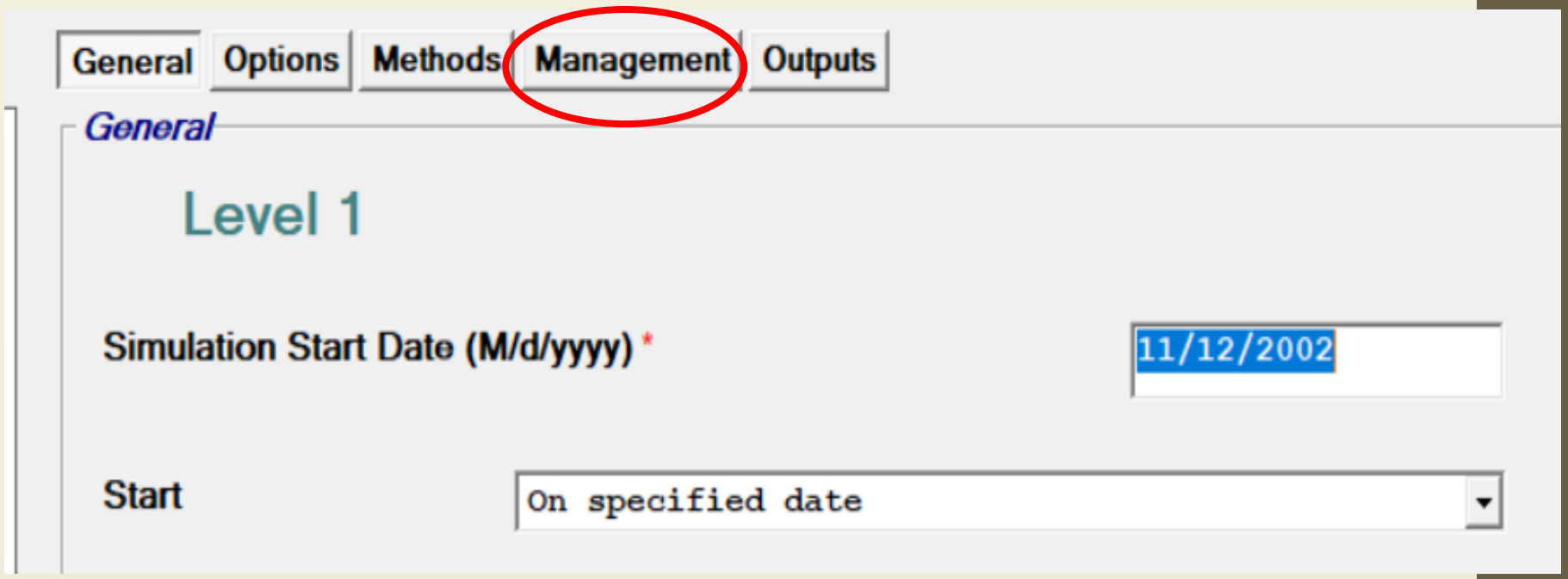
D. Marwa Samy

Optimization
7) Simulation Optioons

(a) simulation date must precede the planting date

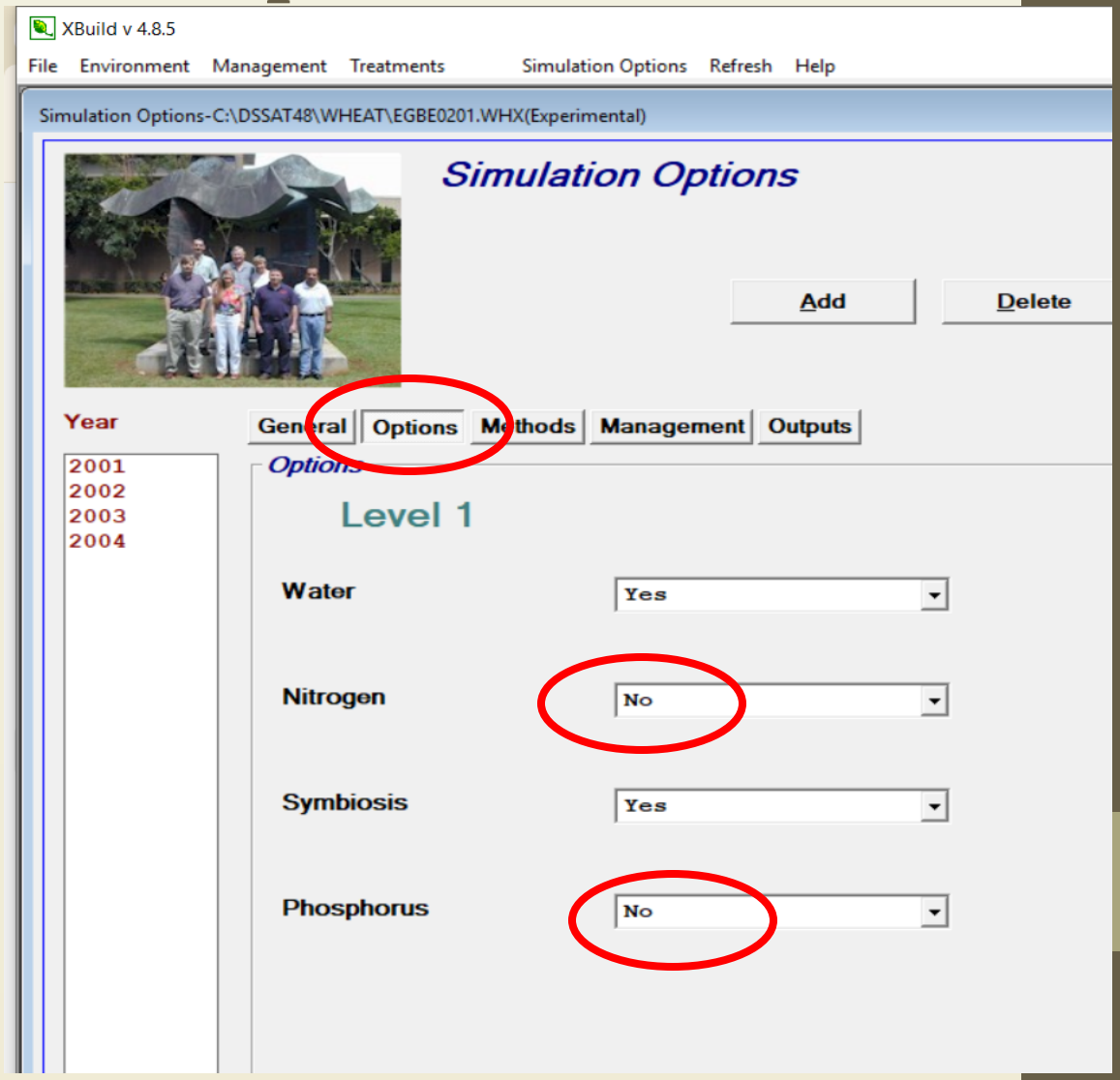


Optimization
7) Simulation → Management



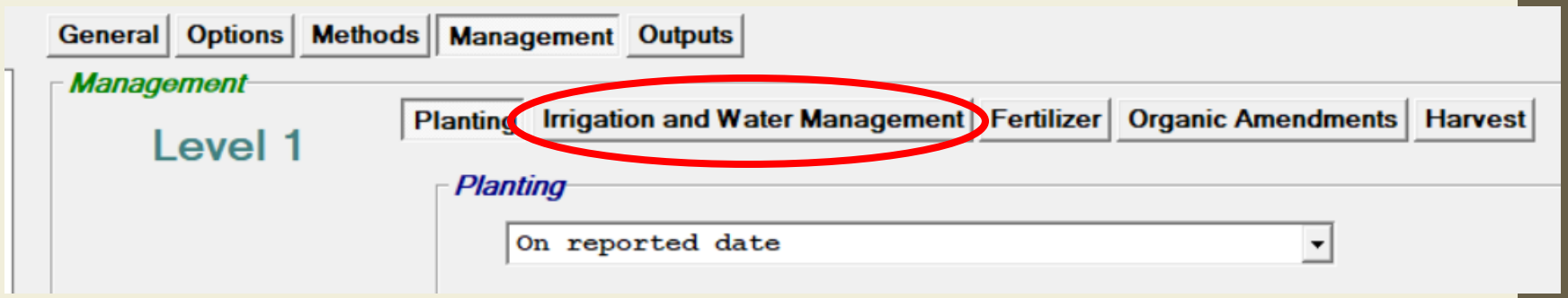
Optimization
7) Simulation → Options

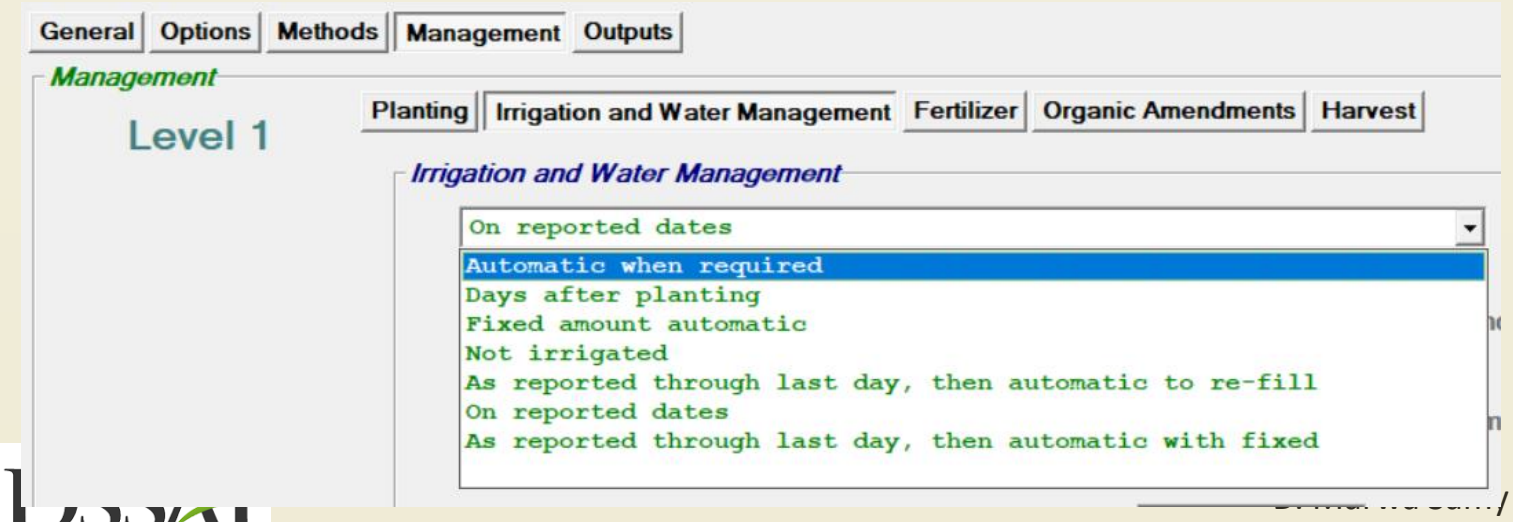
- Stop the si,ulation of fertilizers



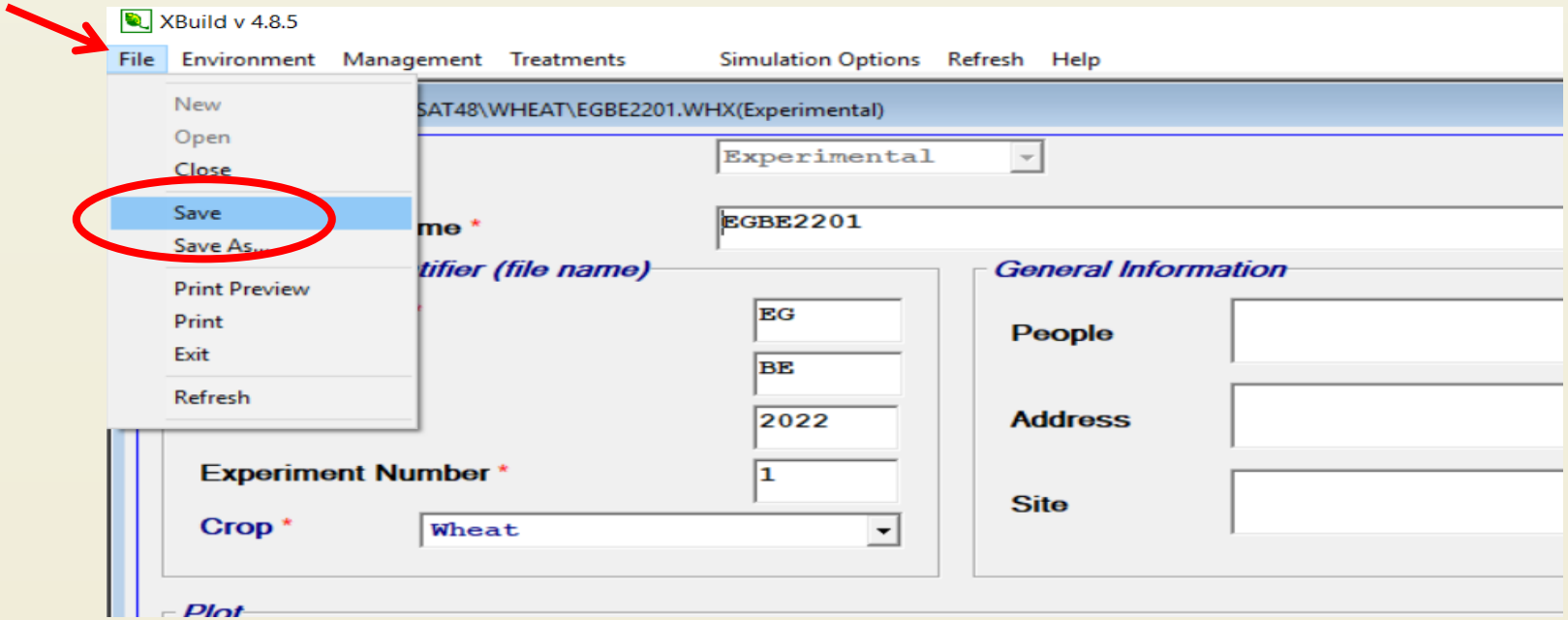
Optimization
8) Simulation → Management

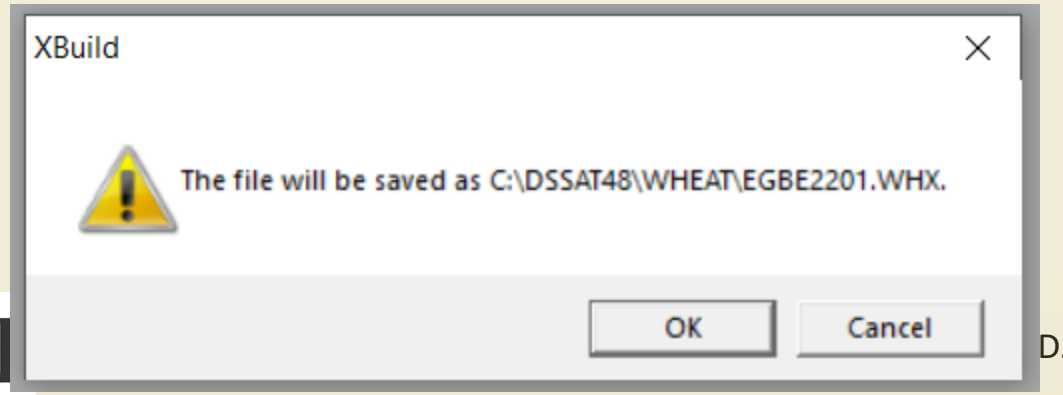
- **Simulation → Management → irrigation and water requirements**





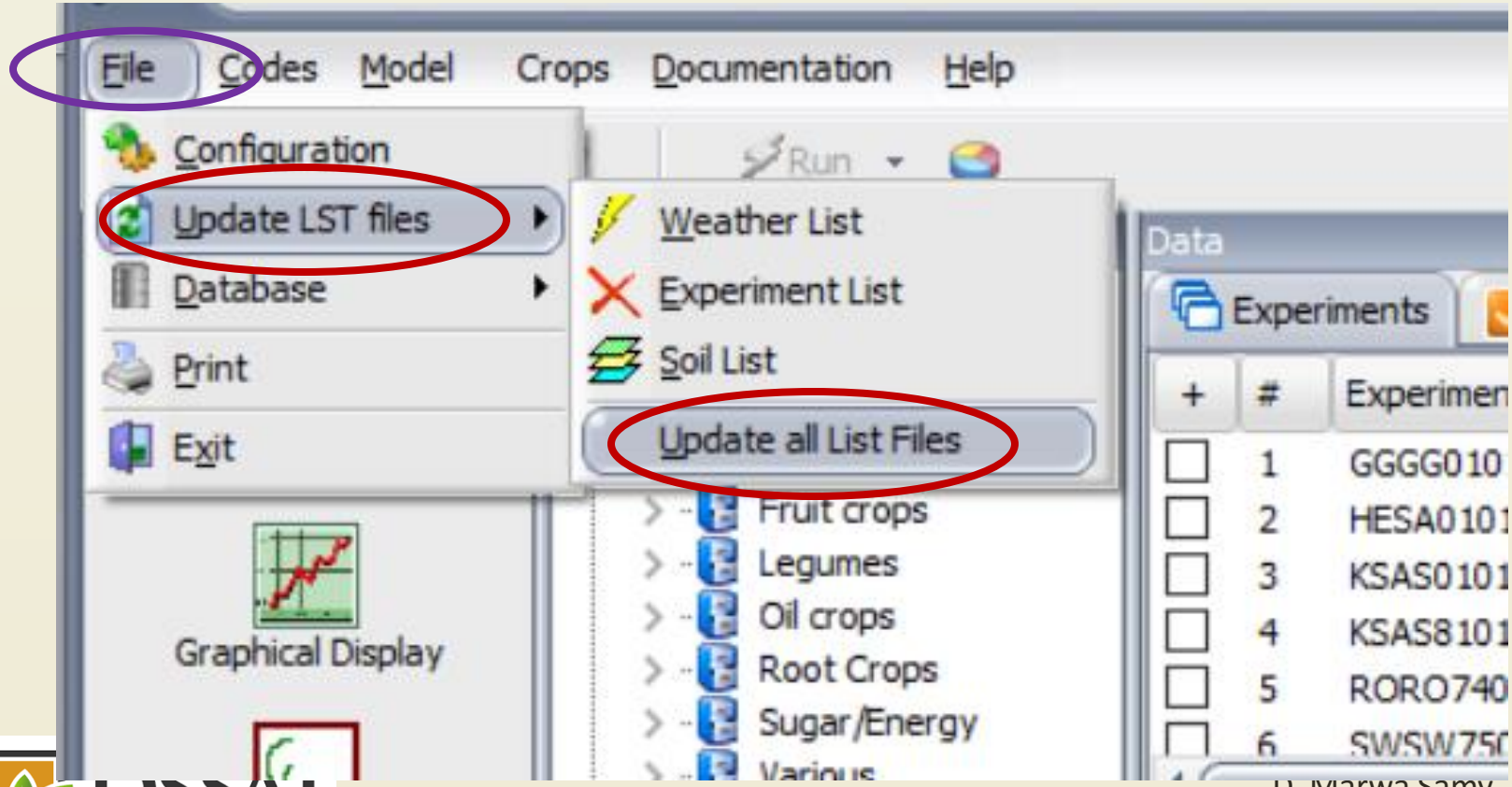
Optimization
9) File → Save





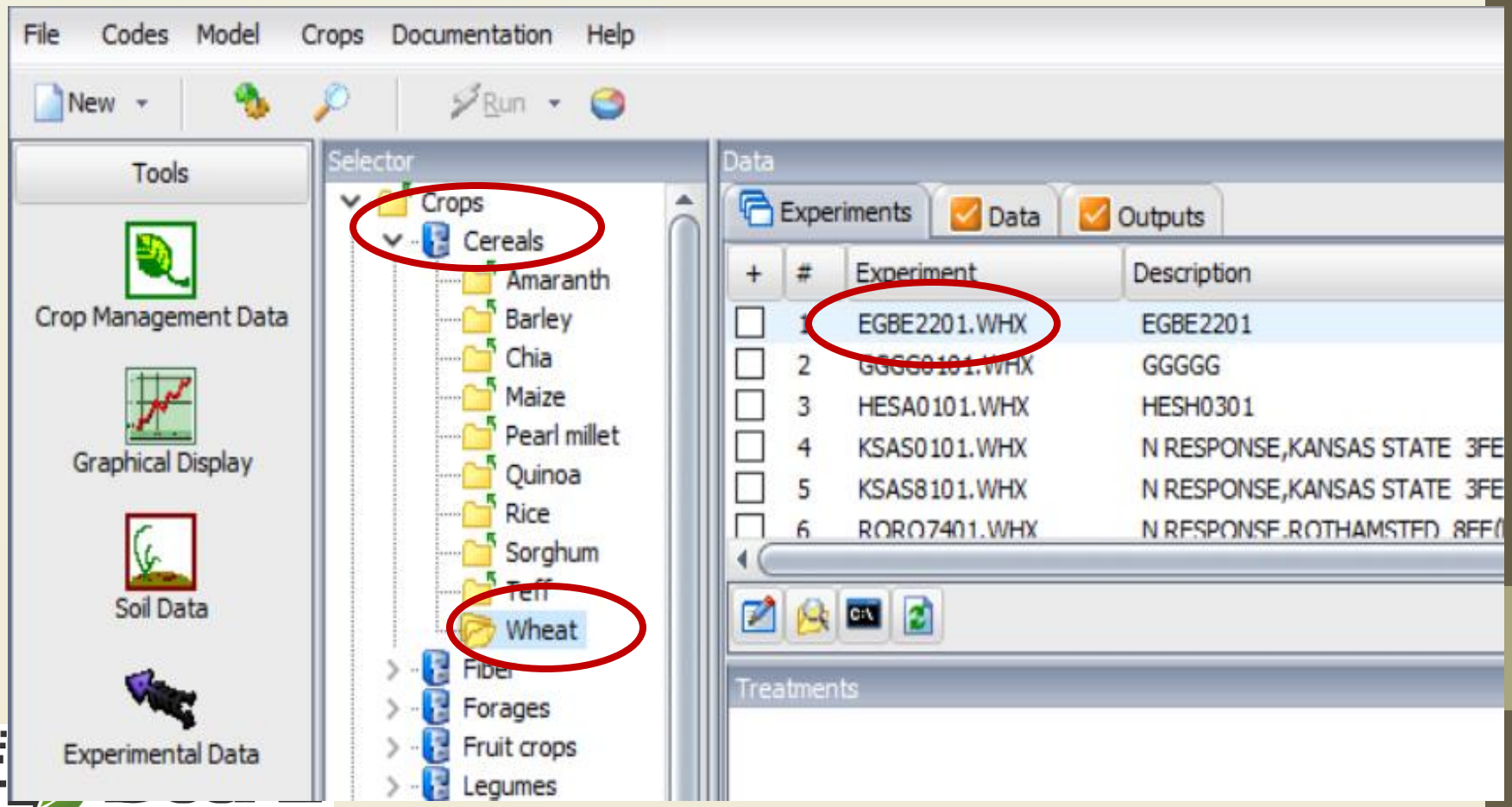
Update your DSSAT

- **File → Update LST files → update all list files**



Find your file

- **Crop → Cereals → Wheat**



Thank you
Any Question